

Tossaporn (Tree) Saengja

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BIO, EDUCATION & AWARD

Five years' experience in research, industry, and teaching. Developed advanced machine learning models and led teams.

Massachusetts Institute of Technology

Masters of Engineering in Computer Science and Engineering 4.8/5.0 2019 – 2020

Bachelor of Science in Computer Science and Engineering 4.8/5.0 2015 – 2019

International Olympiad in Informatics 2013 (IOI), Bronze, 85th among 299 contestants over 77 countries

EXPERIENCE

SiData+, Siriraj Hospital — Data Scientist 2024 – Present

- Collaborating with SCB 10x to develop a medical reasoning Large Language Model (LLM) for general medical purpose.

PreceptorAI, CARIVA — AI Consultant 2023 – Present

- Developed machine learning models in collaboration with Siriraj Radiology Department:
 - a multi-label (muscle, fats) segmentation model (nnUNet, 0.98 Dice) for vertebrae L3 in MRI images (PyTorch).
 - an image-to-text model (Vision-Large Language Model) on chest X-ray images to report findings (PyTorch).
 - a generative text-to-image model (diffusion model) for chest X-ray images (PyTorch, MONAI).
- Led Research on Gindee, a nutrition estimator platform, using Vision-Large Language models.

Vision and Learning Lab, VISTEC — Research Assistant 2023 – 2024

- Under supervision of Prof. Supasorn Suwajanakorn, worked on generative models (e.g. diffusion models).
- Worked on collocation in real images by distilling stable diffusion's prior knowledge (Score Distillation Sampling).
- Worked on local attention masking in CLIP for zero-guidance segmentation.
- Experimented on memorization of diffusion models in CIFAR-10 dataset.

Facebook, Notification Backend — Intern 2017

- Implemented a production pipeline for network effect prediction models, both classification and regression, using HiveQL and FBLeanner Flow within Facebook's machine learning framework.
- Increased 0.5% comments metrics in production with 2 billion active users at the time.

PUBLICATIONS

LUSD: Localized Update Score Distillation for Text-Guided Image Editing

Worameth Chinchuthakun*, Tossaporn Saengja*, Nontawat Tritrong, Pitchaporn Rewatbowornwong, Pramook Khungurn, Supasorn Suwajanakorn. [arXiv:2503.11054](https://arxiv.org/abs/2503.11054) (co-first, under review)

ZeroGuideSeg++: Improving Local Attention Masking in CLIP for Zero-Guidance Segmentation

Pitchaporn Rewatbowornwong, Nattanat Chatthee, Tossaporn Saengja, Ekapol Chuangsuwanich, Supasorn Suwajanakorn. (under review)

GPU Poor's Guide to Radiology Report Generation

Kiartnarin Udomlaksakul, Parinthapat Pengpun, Tossaporn Saengja, Kanyakorn Veerakanjana, Krittamate Tiankanon, Pitikorn Khlaisamniang, Pasit Supholkhan, Amrest Chinkamol, Pubordee Aussavavirojekul, Hirunkul Phimsiri, Tara Sripo, Chiraphat Boonnag, Trongtum Tongdee, Thanongchai Siriapisith, Pairash Saiviroonporn, Jiramet Kinchagawat, Piyalitt Ittichaiwong. [The 23rd Workshop on Biomedical Natural Language Processing and BioNLP Shared Tasks](https://arxiv.org/abs/2408.14541).

ADDITIONAL EXPERIENCE

Mahidol Wittayanusorn High School — Computer Science Teacher 2020 – 2023

- Coached students to represent Thailand in IOI 2022.
- Lectured data science (Python), special topics (Web Development, Blockchain), Computer Olympiad (C++), and programming (Python).
- Advised year-long student projects, mostly machine learning.

Agnos — Technical Co-founder

2019 – 2023

- Developed, with physicians, a diagnosis system (Python, Django) (130k diagnosis records, 50k active users).
- Led Data Science, AI, and Growth teams.
- Raised funds in Shark Tank Thailand S3 for 30M baht.

MIT Media Lab, Human Dynamics — Researcher

2018 – 2020

- Under supervision of Yan Leng, developed a novel optimization framework to learn the underlying network structure and individual marginal benefits in large-scale, real-world datasets.
- Learning Quadratic Games on Large-Scale Networks: Prepared large-scale, real-world datasets for optimization frameworks to learn the underlying network structure and individual marginal benefits.